USB PD CTS ENGINEERING CHANGE NOTICE

NOTICE: Any Company or Companies submitting a USB Power Delivery ECR proposal must be one of the following: a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum. If a group of Companies is submitting an ECR proposal, each company must be either a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum.

SPECIFICATION REVISIONS AND ADDENDA: At any point in time, there shall only be one current version of the USB PD CTS, termed the production version. At the same time, there may also be proposed revisions to the specification's design which are not yet approved and shall be held confidential as deemed necessary by the USB 3.0 and USB 2.0 Promoters and within the Group of Working Committee(s).

PROCEDURES FOR SUBMITTING PROPOSALS: Both members of the USB Implementers Forum as a whole and members of the USB 3.0 and USB 2.0 Promoters may submit requests to revise the USB PD CTS Specification. Such a request may be rejected or may result in a USB PD Engineering Change Notice (ECN), which is the official way USB specifications may be changed.

FORMAT OF PROPOSAL: The originator of a request to alter the USB PD CTS Specification may do so by posting this to the USB Power Delivery Compliance working group for review. Once the proposal has been reviewed by the working group it will be passed to the USB 3.0 and 2.0 Promoters for approval to publish.

RESUBMISSION AND APPEAL: The originator of a request that was not approved can redraft the original request. Rewritten proposal will be treated as a new proposal and will be evaluated using the procedures described above. The originator of a request that was not approved can also submit an appeal to the USB 3.0 and 2.0 Promoters. The appeal must be made in writing and addressed to the Secretary of the USB Implementers Forum.

ABOUT THE ENGINEERING CHANGE REQUEST FORM:
The Purpose of this Engineering Change Request Form is to expedite the review process of the proposal by providing explanations, background information, and examples of the proposed changes at a high level. This form serves as an executive summary to the actual proposal.

STEPS ON HOW TO SUBMIT A USB PD ENGINEERING CHANGE REQUEST:

1) Please fill out the Engineering Change Request Form on the following pages completely:
   a) Detail the names and contact details for each of the ECR contributors
   b) Update the ECR Title
   c) Give a minimum of 2-3 sentences for each description on the form outlining the background to the ECR
2) For each section/table/figure to be updated:
   a) Detail the section number, starting page and figure/table number to be updated as appropriate.
   b) Detail existing text under "From Text"
   c) Detail changed text under "To Text"
3) Save the file as "USB PD CTS 1.0 R 1" followed by the ECR Title as per step 1)b)
4) Post the ECR in the USB PD CTS Documents section under "ECR | New ECRs".
   a) This ECR will then be reviewed by the Power Delivery Compliance Working Group.
   b) Revisions to the ECR originating from the review should be submitted as document revision of the original ECR using "Add new document".

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USB PD CTS ENGINEERING CHANGE NOTICE

Title: UFP VDO TEST
Applied to: USB PD CTS Specification Q3 2026

Brief description of the functional changes proposed:
Adding USB4 UFP VDO check for Speed.

Benefits as a result of the proposed changes:
The vendors will properly define USB4 UFP VDO, and DFPs, as an example, configures USB4 speed properly

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:

An analysis of the hardware implications:

An analysis of the software implications:

An analysis of the compliance testing implications:

An analysis of the Vendor Info File (VIF) implications:

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Actual Change Requested

(a). Section TEST.PD.USB4.EUSB1, Page 310

From Text:

2. The Tester sends a *Discover Identity* Command to the UUT, using SOP. The Tester checks the response. [TEST.PD.USB4.EUSB.1#3]

a. The check fails if the UUT responds with a "Responder NAK".
b. The check fails if the UUT responds with *Not_Supported*
c. If the UUT interrupts the Discovery Process (by sending anything but a *Discover Identity* Response message), the Tester resends *Discover Identity* Command *tVDMBusy* min after it is in PE_SNK_RDY for *nBusyCount* times. The check fails if the Tester has resent *Discover Identity* Command for *nBusyCount* times.
d. The check fails if the UUT responds with a "Responder BUSY" message and the Tester has resent *Discover Identity* Command for *nBusyCount* times every *tVDMBusy* min.
e. The check fails if the UUT does not send "Responder ACK"

To Text:

2. The Tester sends a *Discover Identity* Command to the UUT, using SOP. The Tester checks the response. [TEST.PD.USB4.EUSB.1#3]

a. The check fails if the UUT responds with a "Responder NAK".
b. The check fails if the UUT responds with *Not_Supported*
c. If the UUT interrupts the Discovery Process (by sending anything but a *Discover Identity* Response message), the Tester resends *Discover Identity* Command *tVDMBusy* min after it is in PE_SNK_RDY for *nBusyCount* times. The check fails if the Tester has resent *Discover Identity* Command for *nBusyCount* times.
d. The check fails if the UUT responds with a "Responder BUSY" message and the Tester has resent *Discover Identity* Command for *nBusyCount* times every *tVDMBusy* min.
e. The check fails if the UUT does not send "Responder ACK"
f. The check fails if the UUT UFP VDO:
    i. USB4 Device Capability Bit 27 is not set to 1b
    ii. USB Highest Speed (Bits 2…0) does not match to the VIF USB4_Max_Speed parameter

(b). Section TEST.PD.USB4.EUSB2, Page 313

From Text:

2. The Tester sends a *Discover Identity* Command to the UUT, using SOP. The Tester checks the response. [TEST.PD.USB4.EUSB.2#3]

a. The check fails if the UUT responds with a "Responder NAK".
b. The check fails if the UUT responds with *Not_Supported*
c. If the UUT interrupts the Discovery Process (by sending anything but a *Discover Identity* Response message), the Tester resends *Discover Identity* Command *tVDMBusy* min after it is in PE_SNK_RDY for *nBusyCount* times. The check fails if the Tester has resent *Discover Identity* Command for *nBusyCount* times.

d. The check fails if the UUT responds with a “Responder BUSY” message and the Tester has resent *Discover Identity* Command for *nBusyCount* times every *tVDMBusy* min.
e. The check fails if the UUT does not send “Responder ACK”

To Text:

2. The Tester sends a *Discover Identity* Command to the UUT, using SOP. The Tester checks the response. [TEST.PD.USB4.EUSB.2#3]

a. The check fails if the UUT responds with a “Responder NAK”.
b. The check fails if the UUT responds with *Not_Supported*
c. If the UUT interrupts the Discovery Process (by sending anything but a *Discover Identity* Response message), the Tester resends *Discover Identity* Command *tVDMBusy* min after it is in PE_SNK_RDY for *nBusyCount* times. The check fails if the Tester has resent *Discover Identity* Command for *nBusyCount* times.
d. The check fails if the UUT responds with a “Responder BUSY” message and the Tester has resent *Discover Identity* Command for *nBusyCount* times every *tVDMBusy* min.
e. The check fails if the UUT does not send “Responder ACK”
f. The check fails if the UUT UFP VDO:
    i. USB4 Device Capability (Bit 27) is not set to 1b
    ii. USB Highest Speed (Bits 2…0) does not match to the VIF USB4_Max_Speed parameter